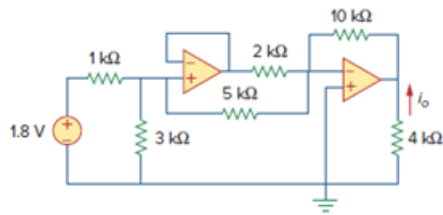


Problem

Calculate i_o in the op amp circuit of Fig. 5.85.

Figure 5.85:



Step-by-step solution

Step 1 of 5

Refer to figure 5.85 given in the textbook.

Redraw the circuit as shown in figure 1.

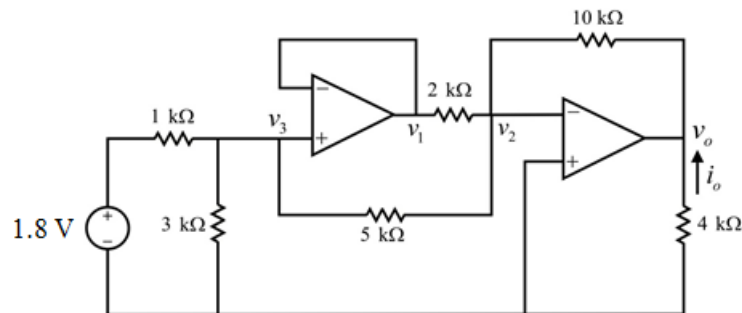


Figure 1

Step 2 of 5

Apply Kirchhoff's current law at node v_3 .

$$\frac{v_3 - 1.8}{1 \text{ k}\Omega} + \frac{v_3}{3 \text{ k}\Omega} + \frac{v_3 - v_2}{5 \text{ k}\Omega} = 0$$

$$\frac{v_3 - 1.8}{1} + \frac{v_3}{3} + \frac{v_3 - v_2}{5} = 0$$

The node voltage v_2 is equal to zero. Since, both inverting and non-inverting terminals of an op-amp is virtually grounded. Therefore, the one end of the resistance of $5 \text{ k}\Omega$ is grounded.

Substitute 0 for v_2 .

$$\frac{v_3 - 1.8}{1} + \frac{v_3}{3} + \frac{v_3}{5} = 0$$

$$v_3 \left(1 + \frac{1}{3} + \frac{1}{5} \right) = 1.8$$

$$v_3 \left(\frac{23}{15} \right) = \frac{18}{10}$$

$$v_3 = \frac{27}{23} \text{ V}$$

Step 3 of 5

Calculate node voltage v_1 .

$$v_1 = v_3$$

Since, both inverting and non-inverting terminals of an op-amp is virtually grounded.

Substitute $\left(\frac{27}{23}\right)$ for v_3 .

$$v_1 = \frac{27}{23} \text{ V}$$

Step 4 of 5

Apply Kirchhoff's current law at node v_2 .

$$\frac{v_2 - v_1}{2 \text{ k}\Omega} + \frac{v_2 - v_3}{5 \text{ k}\Omega} + \frac{v_2 - v_o}{10 \text{ k}\Omega} = 0$$

$$\frac{v_2 - v_1}{2} + \frac{v_2 - v_3}{5} + \frac{v_2 - v_o}{10} = 0$$

Substitute 0 for v_2 , and $\left(\frac{27}{23}\right)$ for v_1 and v_3 .

$$\frac{0 - \left(\frac{27}{23}\right)}{2} + \frac{0 - \left(\frac{27}{23}\right)}{5} + \frac{0 - v_o}{10} = 0$$

$$\frac{v_o}{10} = -\left(\frac{27}{23}\right)\left(\frac{1}{2} + \frac{1}{5}\right)$$

$$v_o = \frac{-270}{23}\left(\frac{7}{10}\right)$$

$$v_o = \frac{-189}{23} \text{ V}$$

Step 5 of 5

Calculate current i_o .

Apply Ohm's law.

$$v_o = -(4 \text{ k}\Omega)i_o$$

$$i_o = \frac{-v_o}{4 \text{ k}\Omega}$$

Substitute $\frac{-189}{23}$ for v_o .

$$i_o = \frac{-\left(\frac{-189}{23}\right)}{4 \text{ k}\Omega}$$

$$= 2.054 \text{ mA}$$

Therefore, the value of current i_o is 2.054 mA.